

Multi-Region Disaster Recovery Plan

Services: EC2, RDS

Description:

Implement a disaster recovery architecture where applications and databases are deployed across two AWS regions. Configure cross-region database replication to ensure business continuity and data availability during regional outages.

Skills: High Availability, Disaster Recovery Planning, Cross-Region Replication, Database Management, Business Continuity.

Definition: The Multi-Region Disaster Recovery Plan is designed to provide a resilient and highly available infrastructure by maintaining application and database resources in two separate AWS regions. The primary region hosts the production workload, while the secondary region acts as a disaster recovery (DR) environment.

The solution uses Amazon EC2 instances for application hosting and Amazon RDS cross-region replication for database synchronization. In the event of a failure in the primary region, services can be restored from the secondary region, minimizing downtime and reducing data loss.

Scope of the Project:

The scope of this project is to design and implement a disaster recovery architecture using AWS services. The project focuses on ensuring application availability and database protection through cross-region deployment and replication.

This project covers:

- Primary Region Deployment: Deploy application servers and databases in the primary AWS region.
- Disaster Recovery Region Deployment: Deploy standby infrastructure in a secondary AWS region.
- Cross-Region Database Replication: Replicate data from the primary RDS database to a secondary region using Read Replicas.
- Disaster Recovery Testing: Simulate failures and validate recovery procedures.
- Business Continuity: Ensure applications remain available during regional outages.

Methodology:

1. Cross-Region Replication Method – Best Method

- Uses AWS services: EC2 and RDS.
- Provides continuous database replication between regions.
- Reduces Recovery Point Objective (RPO).
- Suitable for production and enterprise workloads.
- Ensures business continuity during disasters.
- Requires database replication configuration and monitoring.

2. Backup and Restore Method – Alternative Method

- Uses RDS automated backups and snapshots.
- Lower infrastructure cost.
- Simpler implementation.
- Suitable for development and testing environments.
- Recovery process requires manual database restoration.
- Recovery time is higher compared to replication-based DR.

AWS Services used in the Project

1. Amazon EC2 (Elastic Compute Cloud)

Definition: Amazon EC2 is a web service that provides secure and resizable virtual servers in the cloud, allowing users to deploy applications and workloads on demand.

Purpose: Provides application servers in both primary and disaster recovery regions.

Role: Hosts web applications and services that user's access.

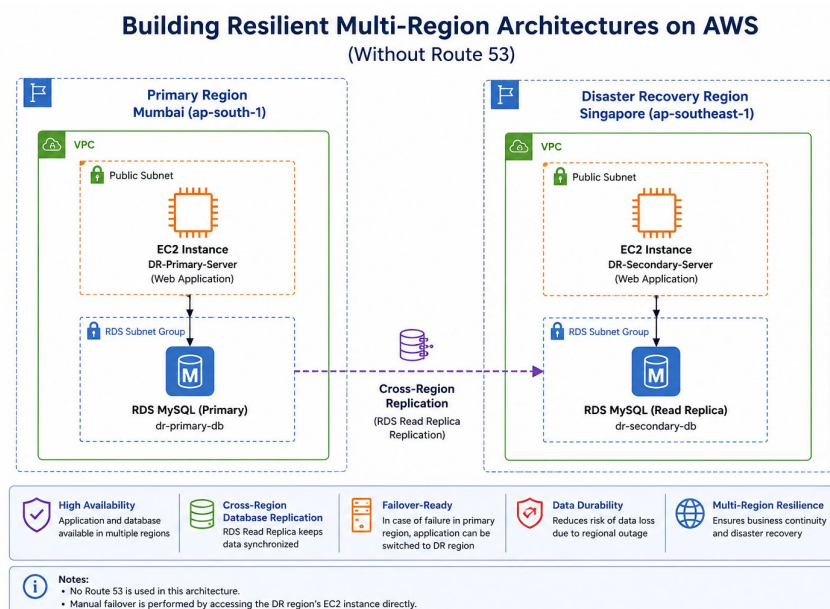
2. Amazon RDS (Relational Database Service)

Definition: Amazon RDS is a managed database service that simplifies database administration, backups, patching, monitoring, and replication.

Purpose: Stores application data and provides database replication capabilities.

Role: Maintains the primary database and replicates data to the disaster recovery region through Read Replicas.

Architecture Diagram

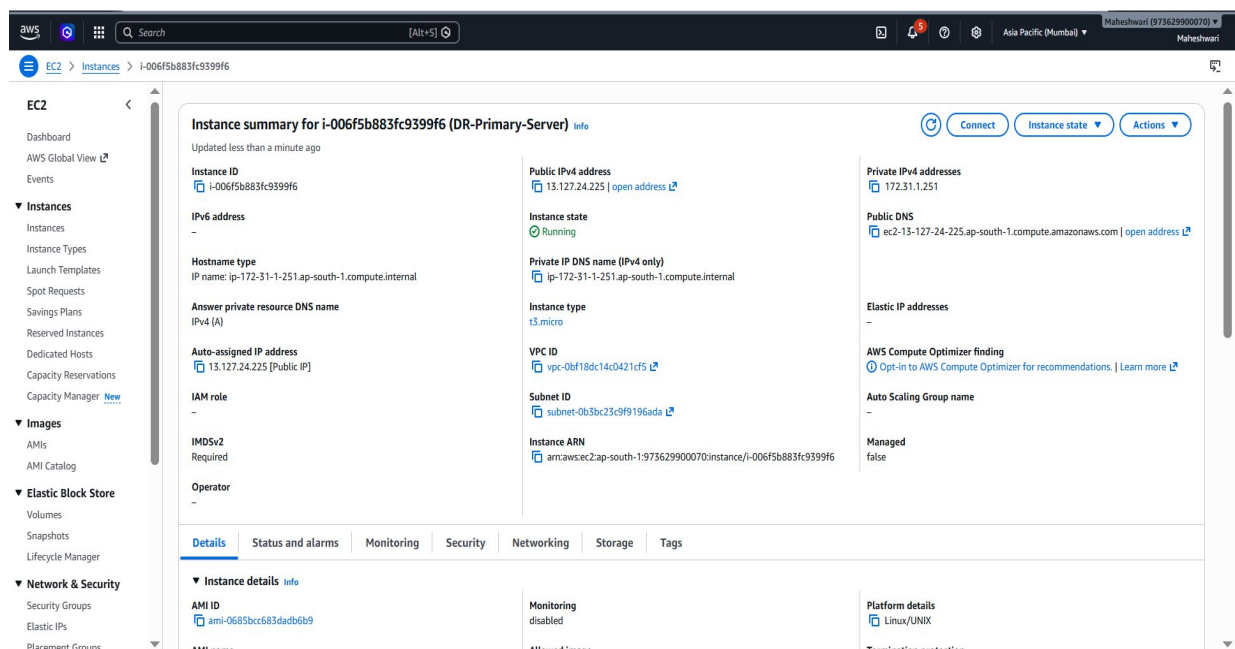


Implementation Steps to Do the Project

Step 1: Launch Primary EC2 Instance

- Go to AWS Console → EC2 → Launch Instances.
- Choose Amazon Linux 2023.
- Configure:
 - Default VPC
 - Public Subnet
 - Auto Assign Public IP
- Configure Security Group:
 - SSH (22)
 - HTTP (80)
- Launch Instance.
- Name:

DR-Primary-Server



Step 2: Install Web Server

Connect to EC2 using SSH.

Install Apache:

```
sudo yum update -y
```

```
sudo yum install httpd -y
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

Create Web Page:

```
echo "Primary Region Website" | sudo tee /var/www/html/index.html
```

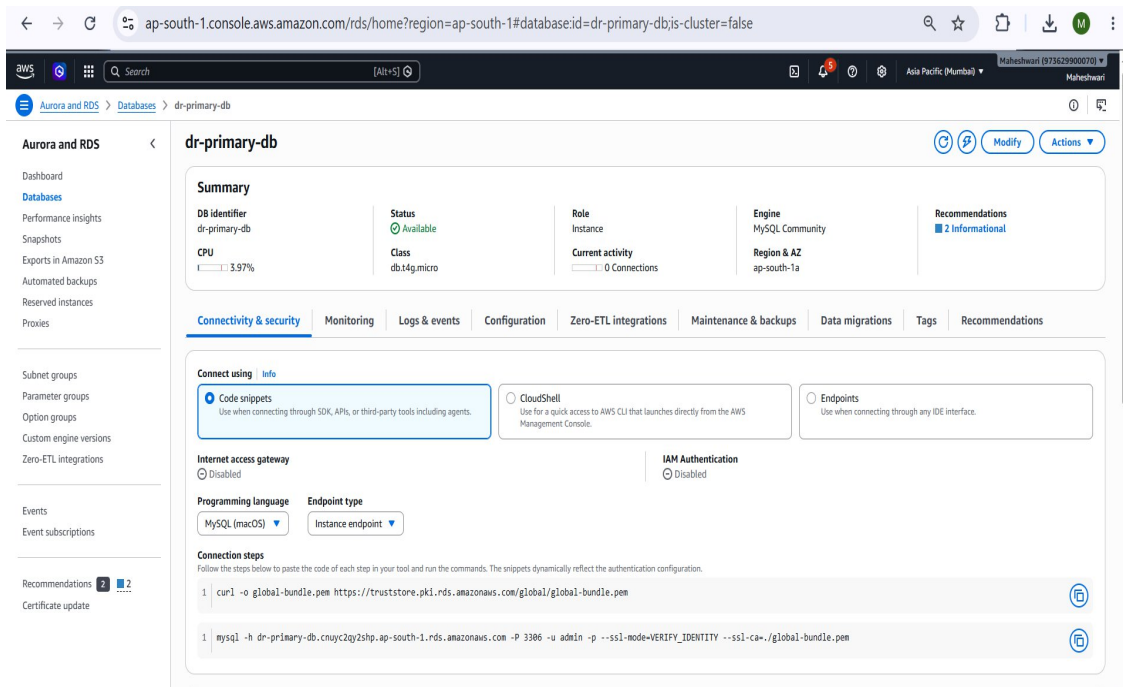
Verify Website:

<http://Primary-EC2-Public-IP>



Step 3: Create Primary RDS Database

- Go to AWS Console → RDS → Create Database.
- Engine: MySQL
- DB Identifier: dr-primary-db.
- Username: admin
- Storage: 20 GB
- Enable Automated Backups.
- Create Database.

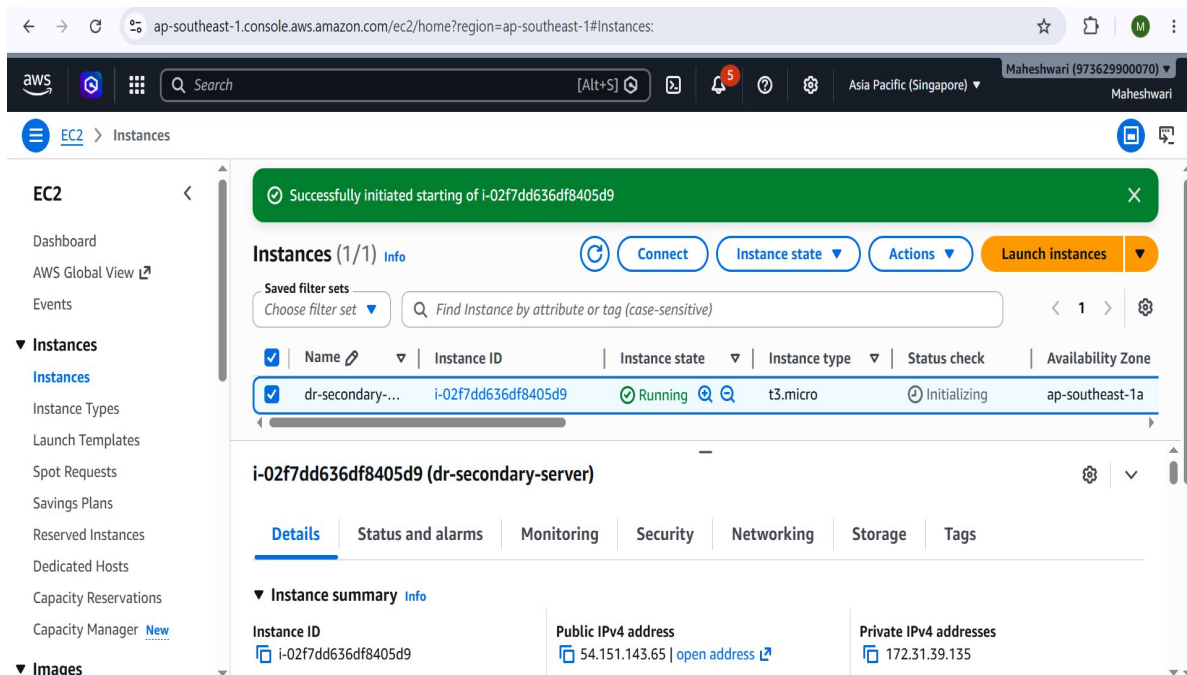


Step 4: Launch Disaster Recovery EC2 Instance

Switch Region: Singapore ap-southeast-1

Launch another EC2 Instance.

Name: DR-Secondary-Server



Install Apache.

Create Web Page:

echo "Disaster Recovery Region Website" | sudo tee /var/www/html/index.html

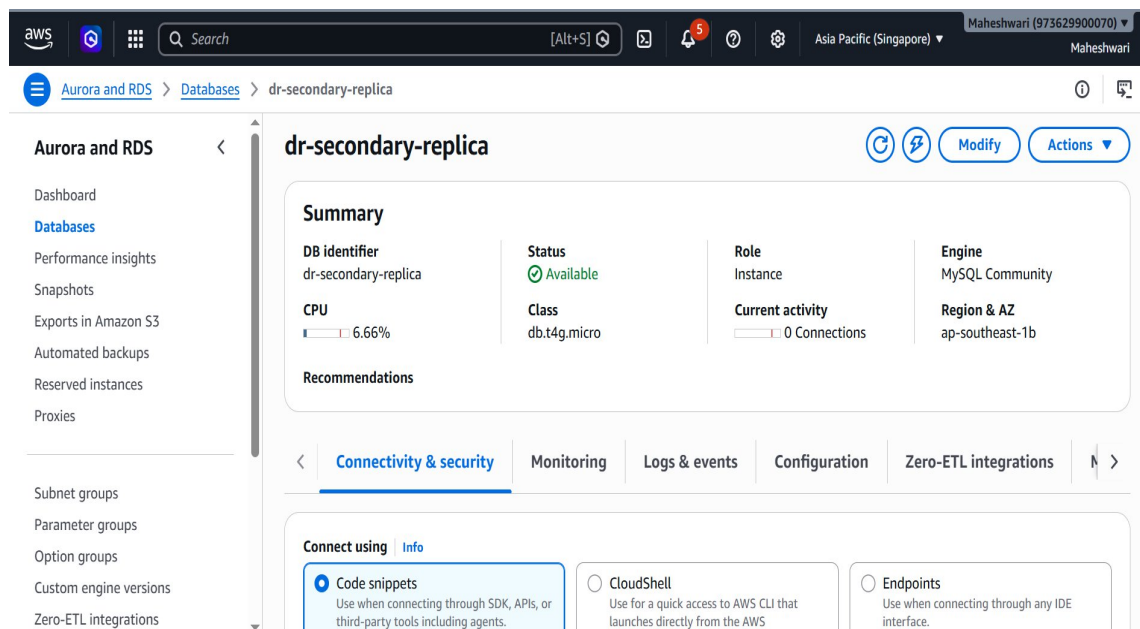
Verify Website:

<http://Secondary-EC2-Public-IP>



Step 5: Configure Cross-Region Read Replica

- Open RDS Console.
- Select: Source-dr-primary-db.
- Destination Region = Singapore (ap-southeast-1)
- DB Identifier = dr-secondary-replica
- Actions → Create Read Replica.
- Destination Region: Singapore
- Create Replica.
- AWS creates: dr-secondary-replica



Step 6: Verify Database Replication

Connect to Primary Database.

Create Database:

```
CREATE DATABASE drtest;
```

```
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i-006f5b883fc9399f...

[root@ip-172-31-1-251 ec2-user]# mysql -h dr-primary-db.cnuy2qy2shp.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 84
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| drprojectdb |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.027 sec)

MySQL [(none)]> CREATE DATABASE drtest
-> SHOW DATABASES;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'SHOW DATABASES' at line 2
MySQL [(none)]> CREATE DATABASE drtest;
Query OK, 1 row affected (0.030 sec)

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| drprojectdb |
| drtest |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.001 sec)

MySQL [(none)]>
```

```
ERROR 1045 (28000): Access denied for user 'admin'@'172.31.1.251' (using password: NO)
[root@ip-172-31-1-251 ec2-user]# mysql -h dr-primary-db.cnuy2qy2shp.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
ERROR 1045 (28000): Access denied for user 'admin'@'172.31.1.251' (using password: YES)
[root@ip-172-31-1-251 ec2-user]# ^C
[root@ip-172-31-1-251 ec2-user]# mysql -h dr-primary-db.cnuy2qy2shp.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
ERROR 1045 (28000): Access denied for user 'admin'@'172.31.1.251' (using password: YES)
[root@ip-172-31-1-251 ec2-user]# mysql -h dr-primary-db.cnuy2qy2shp.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
ERROR 1045 (28000): Access denied for user 'admin'@'172.31.1.251' (using password: YES)
[root@ip-172-31-1-251 ec2-user]# mysql -h dr-primary-db.cnuy2qy2shp.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 280
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> USE drtest;
Database changed
MySQL [drtest]>
MySQL [drtest]> CREATE TABLE employees(
  -> id INT,
  -> name VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.060 sec)

MySQL [drtest]> INSERT INTO employees VALUES(1,'Maheshwari');
Query OK, 1 row affected (0.010 sec)

MySQL [drtest]> SELECT * FROM employees;
+-----+-----+
| id | name |
+-----+-----+
| 1 | Maheshwari |
+-----+-----+
1 row in set (0.000 sec)

MySQL [drtest]>
```

i-006f5b883fc9399f6 (DR-Primary-Server)
PublicIPs: 13.206.68.234 PrivateIPs: 172.31.1.251

Connect to Replica Database.

Run:

SHOW DATABASES

Verify: drtest

appears in the replica.

```
aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Singapore) Maheshwari (973629900070) Maheshwari

[root@ip-172-31-39-135 ec2-user]# mysql -h dr-secondary-db.czi6k24g6rju.ap-southeast-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 15
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| drprojectdb |
| drtest |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.069 sec)

MySQL [(none)]> USE drtest;
```

i-02f7dd636df8405d9 (dr-secondary-server)

PublicIPs: 13.212.14.16 PrivateIPs: 172.31.39.135



```
aws [Search] [Alt+S] Asia Pacific (Singapore) Maheshwari (973629900070) Maheshwari

+-----+
| drprojectdb |
| drtest |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.001 sec)

MySQL [drprojectdb]> USE drtest;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [drtest]> SELECT * FROM employees;
+-----+
| id | name |
+-----+
| 1 | Maheshwari |
+-----+
1 row in set (0.002 sec)

MySQL [drtest]>
```

i-02f7dd636df8405d9 (dr-secondary-server)

PublicIPs: 13.212.14.16 PrivateIPs: 172.31.39.135



Step 7: Simulate Disaster Recovery

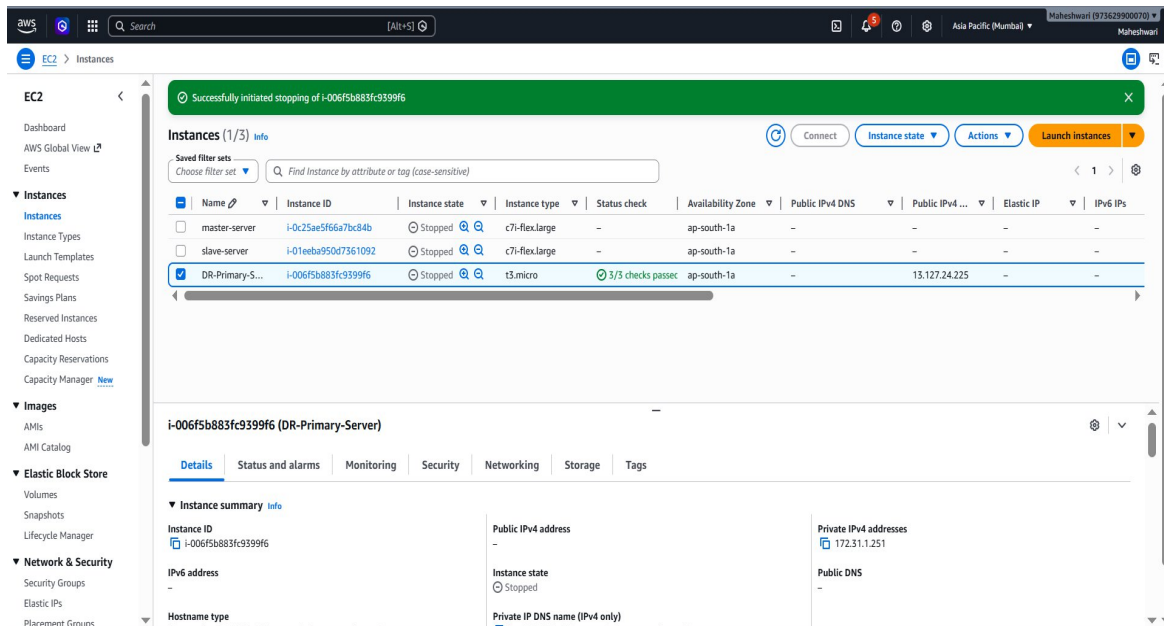
Stop the Primary EC2 Instance.

Go to:

EC2 → Instances

Select: DR-Primary-Server

Click: **Instance State → Stop**



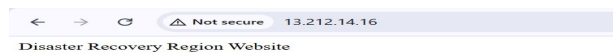
Step 8: Verify Disaster Recovery Environment

Access Secondary Region Website.

<http://Secondary-EC2-Public-IP>

Verify:

Disaster Recovery Region Website



Application remains available from the DR region.

Step 9: Validate Database Availability

Verify:

- Replica database is available.
- Application can connect to the DR database.
- Recovery environment is operational.

```
aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Singapore) Maheshwari (973629900070) Maheshwari
[root@ip-172-31-39-135 ec2-user]# mysql -h dr-secondary-db.czi6k24g6rju.ap-southeast-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 15
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| drprojectdb |
| drtest |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.069 sec)

MySQL [(none)]> USE drtest;
```

i-02f7dd636df8405d9 (dr-secondary-server)

PublicIPs: 13.212.14.16 PrivateIPs: 172.31.39.135

```
aws [Search] [Alt+S] Asia Pacific (Singapore) Maheshwari (973629900070) Maheshwari
+-----+
| drprojectdb |
| drtest |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.001 sec)

MySQL [drprojectdb]> USE drtest;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [drtest]> SELECT * FROM employees;
+----+-----+
| id | name |
+----+-----+
| 1 | Maheshwari |
+----+-----+
1 row in set (0.002 sec)

MySQL [drtest]>
```

i-02f7dd636df8405d9 (dr-secondary-server)

PublicIPs: 13.212.14.16 PrivateIPs: 172.31.39.135

Troubleshooting

Issue 1: Read Replica Creation Failed

The read replica may fail to create if automated backups are disabled.

Solution:

- Enable Automated Backups on the primary database.
- Retry Read Replica creation.

Issue 2: Replication Delay

Database changes may not immediately appear in the replica.

Solution:

- Wait a few minutes for synchronization.
- Verify replication status from the RDS console.

Issue 3: EC2 Website Not Accessible

The website may not load due to security group restrictions.

Solution:

- Verify HTTP Port 80 is allowed.
- Ensure Apache service is running.

```
sudo systemctl status httpd
```

Issue 4: Database Connection Failed

Application cannot connect to RDS.

Solution:

- Verify RDS security group.
- Check endpoint configuration.
- Confirm database credentials.

Conclusion

The Multi-Region Disaster Recovery Plan provides a reliable and scalable solution for ensuring business continuity across AWS regions. By deploying application servers in multiple regions and enabling Amazon RDS cross-region replication, organizations can minimize downtime and reduce the risk of data loss during regional failures.

The architecture improves availability, supports disaster recovery objectives, and demonstrates real-world cloud resilience practices. The solution is suitable for enterprise workloads requiring high availability, business continuity, and disaster preparedness.